

Computer Networks

Lecture 9: Data Link Layer

Content



- **Parity Check**

Question

A block code has the following valid codewords: $C = \{000000, 001111, 110011, 111100\}$

1. Find the minimum Hamming distance ($d_{\min}$) of the code.
2. Determine the maximum number of errors that can be detected.
3. Determine the maximum number of errors that can be corrected.
4. A received word is **110111**.
 - (a) Detect whether an error has occurred.
 - (b) If possible, correct the error and identify the transmitted codeword.

Linear Block Code

In general block codes, the sender and receiver need a lookup table to map:

- datawords $\rightarrow$ codewords
- received codewords $\rightarrow$ nearest valid codewords

As k and n grow, the table size becomes huge.

In a linear block code, the exclusive OR (XOR) of any two valid codewords creates another valid codeword.

Linear Block Code

Minimum Hamming Code

The minimum Hamming distance is the number of 1s in the nonzero valid codeword with the smallest number of 1s.

the most familiar error-detecting code is the simple parity-check code. In this code, a k -bit dataword is changed to an n -bit codeword where $n = k + 1$. The extra bit, called the parity bit, is selected to make the total number of 1s in the codeword even.

Parity

Value of parity bit is such that character has even (even parity) or odd (odd parity) number of ones

Data							Parity
1	0	0	1	1	1	0	0
0	0	1	1	1	0	0	1
1	1	1	0	1	1	1	0

Data							VRC
1	0	0	1	1	1	0	0
0	0	1	1	1	0	0	1
1	1	1	0	1	1	1	0
1	1	1	0	1	1	1	0
0	0	1	1	1	0	0	1
0	0	1	0	1	0	1	1
0	0	1	1	1	0	0	1
1	0	0	0	1	1	1	0

LRC

If performing even parity, add a parity bit such that an even number of 1s are maintained.

If performing odd parity, add a parity bit such that an odd number of 1s are maintained.

Cyclic Codes

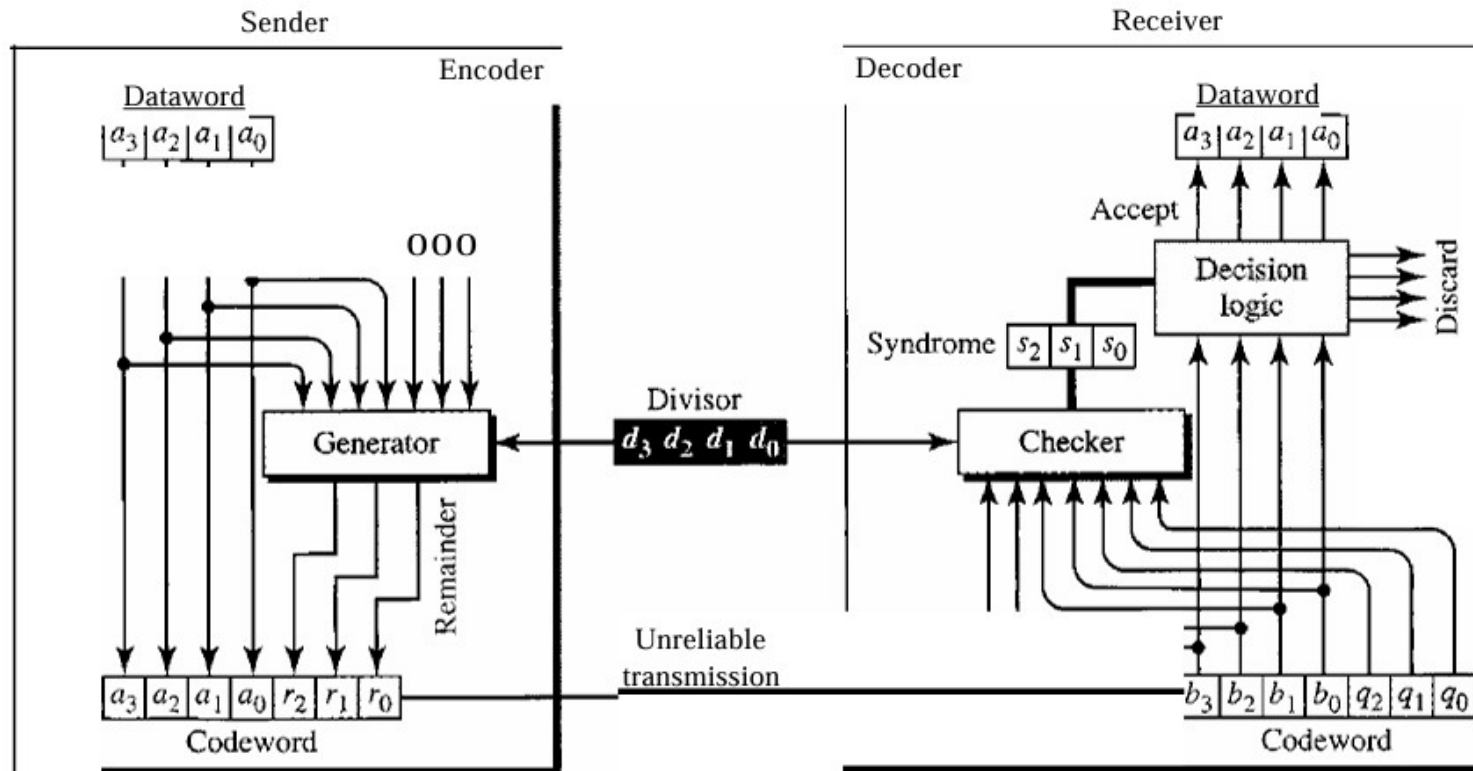
Cyclic codes are special linear block codes with one extra property. In a cyclic code, if a codeword is cyclically shifted (rotated), the result is another codeword

For example

If 1011000 is a codeword and we cyclically left-shift, then 0110001 is also a codeword.

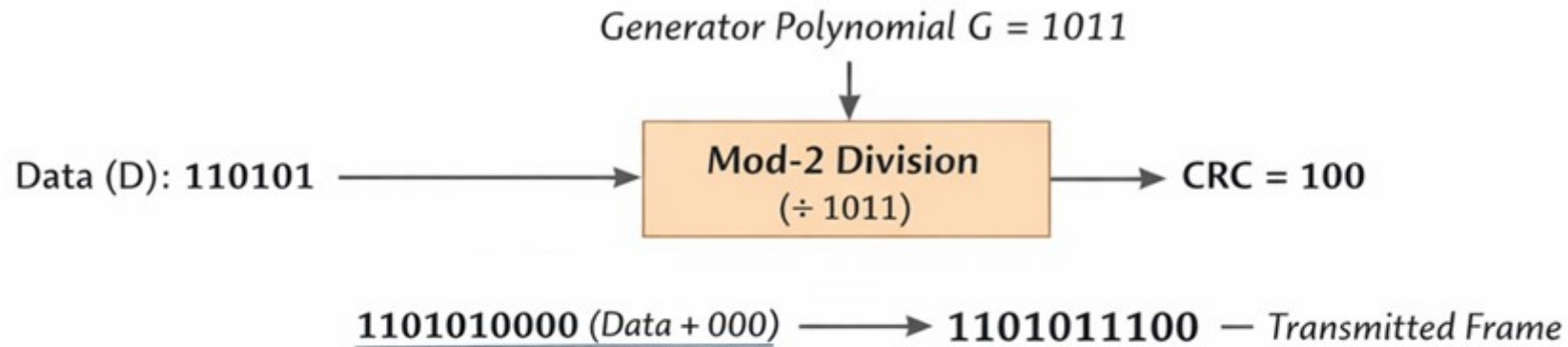
Cyclic Redundancy Check

Cyclic Redundancy Check (CRC) is a robust error-detecting code used in computer networks to identify accidental data changes during transmission.



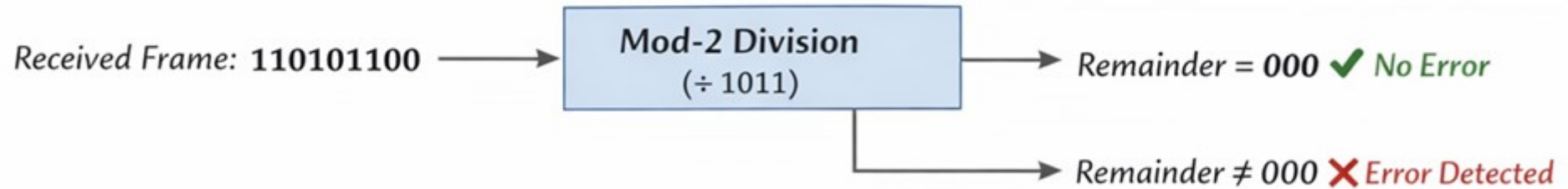
Cyclic Redundancy Check: Example

CRC Encoder



$$1101010000 \div 1011 \rightarrow \text{Remainder: } 100$$

CRC Decoder



$$1101011100 \div 1011 \rightarrow \text{Remainder: } 000 \text{ ✓ Frame Accepted}$$